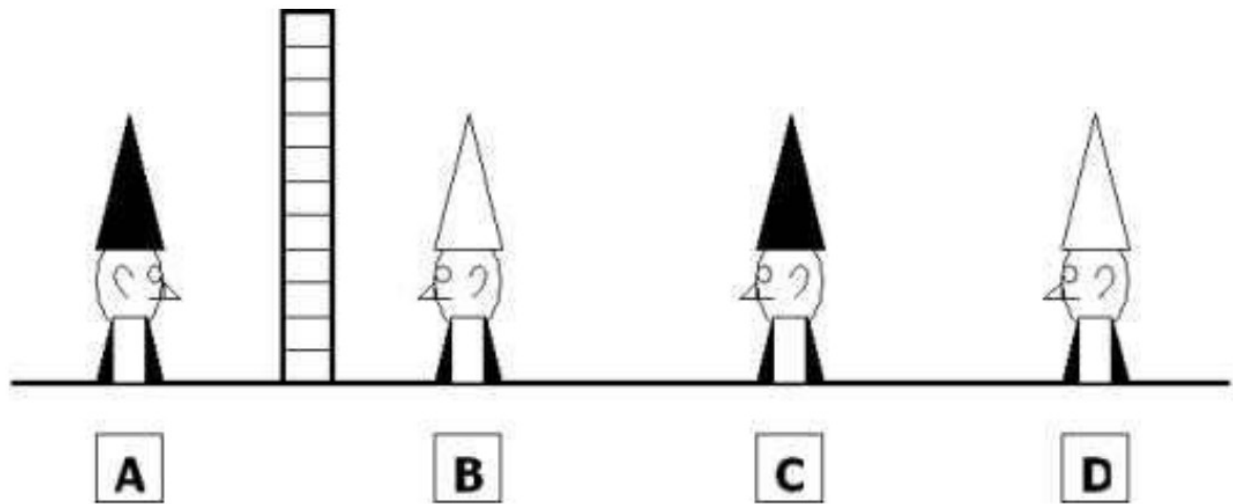


Exercise 30: Four Hats



- Shown above are 4 men buried up to their necks in the ground. They can't move, so can only look forward.
- Between A and B is a brick wall which can not be seen through. They know that between them are 4 hats, 2 x black and 2 x white, but they do not know which color they are wearing.
- In order to avoid being shot one of them must call out to the executioner the color of their hat. If they get it wrong, everyone will be shot.
- They are not allowed to talk to each other and have 10 minutes to fathom it out.
- After 1 minute:
 - Q Which one of them calls out?
 - Why is he 100% certain of the color of his hat?
 - This is not a trick question.
- There are no outside influences nor other ways of communicating. They cannot move and are buried in a straight line. So A & B can only see their respective sides of the brick wall, C can see B and D can see B & C.

Answer Key:

Consider what D can see, he can see 2 hats in front of him. He knows that there are 2 black hats and 2 white hats. If he could see 2 white hats he would know that his hat was black, similarly if he could see 2 black hats he would know his hat was white. Either way he would immediately announce the color of his hat thus saving his own life.

Consider C, if he were to think about the above for D. He would know that if both his hat and the hat in front of him were the same colour, D would call out. Since D, in this case remains silent, C would know from this that his own hat must be different to B's which they can both see. Therefore C would call out that his hat was indeed black.